

Statistics - Measure of Dispersion

① Variance

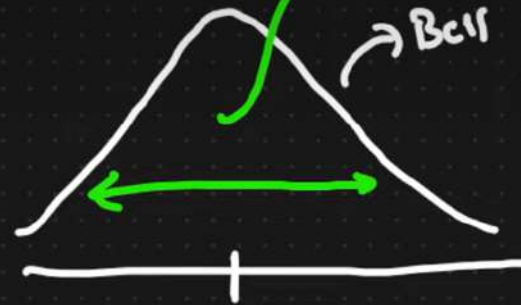
② Standard deviation

→ Spread

Variance, Standard Deviation

→ Bell Curve

Normal Distribution



Mean, Median, Mode

== == ==



Measure of Central Tendency



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30°C Partly sunny

Population (N)

$$\mu = \sum_{i=1}^N \frac{(x_i)}{N}$$

Sample (n)

$$\bar{x} = \sum_{i=1}^n \frac{(x_i)}{n}$$

$$\sigma^2 = \sum_{i=1}^N \frac{(x_i - \mu)^2}{N}$$


$$s^2 = \sum_{i=1}^n \frac{(x_i - \bar{x})^2}{\boxed{n-1}}$$




x	n	$(x - \bar{x})$	$(x - \bar{x})^2$
1	283	-1.83	3.34
2	283	-0.83	0.6889
2	283	-0.83	0.6889
3	283	0.17	0.03
4	283	1.17	1.37
5	283	2.17	4.71
$\bar{x} = 283$			10.84

$$= \frac{17}{6}$$

$S^2 = \frac{10.84}{5} = 2.168$ { Spread of the data }
 $\downarrow$
 Sample Variance

$S.d = \sqrt{\text{Variance}}$
 $= \sqrt{2.168}$



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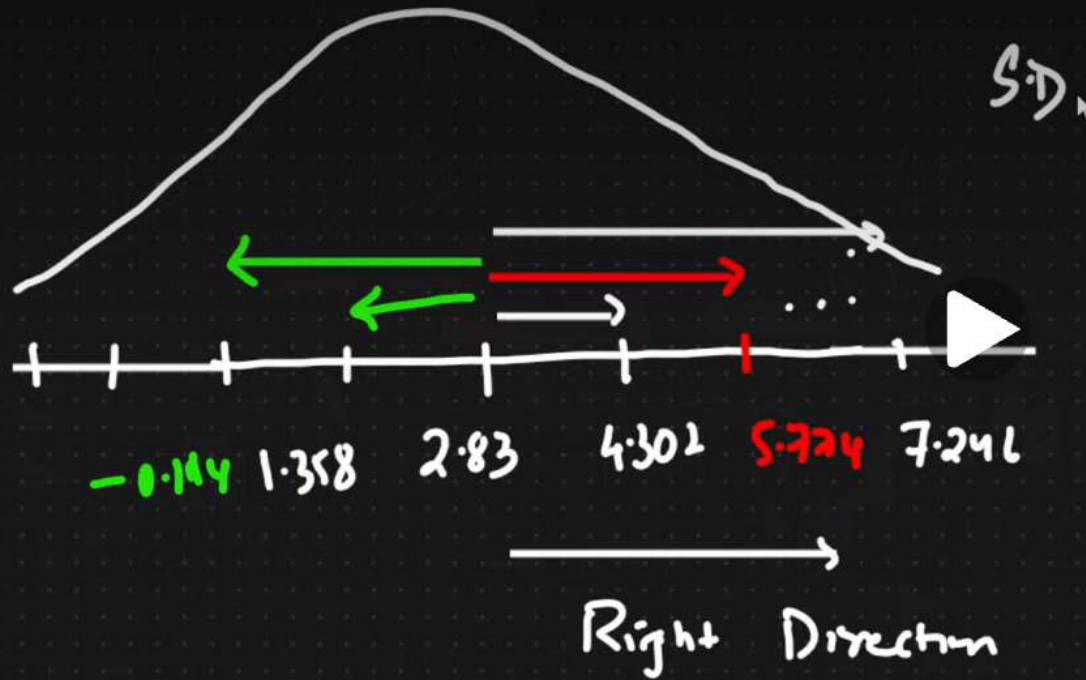
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Dataset

$$\bar{x} = 2.83$$

$$s = 1.472$$



$$\begin{array}{r} 1 \ 1 \\ 2.83 \\ 1.472 \\ \hline 4.302 \\ 1.472 \\ \hline 5.774 \\ 1.472 \\ \hline 7.246 \end{array}$$

$$\begin{array}{r} 2.83 \\ 1.472 \\ \hline 1.358 \end{array}$$



Variance $\rightarrow$ Big Number

S.d $\rightarrow$ "

Spread $\uparrow\uparrow$



Variance $\rightarrow$ Small Number

S.d $\rightarrow$ "

